Notes 3

1. What is a graphical user interface (GUI)?

A GUI is a way to use a computer with windows, icons, menus, and a mouse instead of typing commands.

2. What is a desktop environment?

A desktop environment is the graphical workspace we use on a computer. It includes the desktop, taskbar, windows, settings, and applications. GNOME is an example.

3. What is the command line interface (CLI)?

A CLI is a way to interact with a computer by typing commands instead of clicking icons.

4. How do I access the command line interface (CLI)?

I can open the Terminal application from the applications menu or use the terminal shortcut Ctrl + Alt + T in Ubuntu.

5. What is a virtual console?

A virtual console is a text-based screen where users can log in and enter Linux commands without using the graphical desktop.

6. What is a terminal emulator?

A terminal emulator is an application that lets me type and run commands inside the graphical desktop. Examples include GNOME Terminal and Tilix.

7. What is bash?

Bash stands for Bourne Again Shell. It is a command-line shell that reads and runs commands. It can also run scripts to automate tasks.

8. What is the shell prompt?

The shell prompt is the text that shows the terminal is ready for a command. It usually displays the username, computer name, and current directory.

Example:
`aryan@AryanVM:~$`

Part 2: Commands - Definition, Usage, and Examples

1. clear

Definition: Clears the terminal screen.

Usage: Makes the terminal easier to read.

Example:
```
clear
```

2. echo

Definition: Displays text or variable values in the terminal.

Usage: Prints messages on the screen.

Example:
```
echo "Hello World"
```

3. date

Definition: Displays or sets the system date and time.

Usage: Checks the current date and time.

Examples:

- `date`
- `date --rfc-3339=ns`

4. free

Definition: Displays information about used and available memory.

Usage: Checks RAM and swap memory.

Examples:

- `free`
- `free -h`
- `free -h -L`

5. uname

Definition: Displays information about the operating system and kernel.

Usage: Checks the Linux kernel name, release, and version.

Examples:

- `uname`
- `uname -a`
- `uname -s -r -v -o`

6. history

Definition: Displays previously entered terminal commands.

Usage: Helps me find commands I used earlier.

Examples:

- `history`
- `history 10`

7. man

Definition: Opens the manual page for a command.

Usage: Helps me learn command options and how to use them.

Examples:

- `man free`
- `man uname`

8. tldr

Definition: Shows a short and simple help page for a command.

Usage: Helps me learn commands quickly with examples.

Examples:

- `tldr ls`
- `tldr tar`

9. cheat

Definition: Displays cheat sheets with examples of commands.

Usage: Helps me remember useful Linux commands.

Examples:

- `cheat ls`
- `cheat tar`

10. hostname

Definition: Displays or changes the computer's hostname.

Usage: Checks the computer's name.

Examples:

- `hostname`
- `hostname -I`

11. df

Definition: Shows available and used disk space on file systems.

Usage: Checks how much storage space is available.

Examples:

- `df`
- `df -h`

12. du

Definition: Shows how much disk space files and folders use.

Usage: Checks the size of directories.

Examples:

- `du -h`
- `du -sh ~/Downloads`

13. figlet

Definition: Displays text using large ASCII art letters.

Usage: Creates large text banners in the terminal.

Examples:

- `figlet Hello`
- `figlet Linux`

Conclusion

In this lesson, I learned the difference between GUI and CLI. I also learned how to use the GNOME desktop, terminal emulator, Bash shell, and Linux commands. The `man` command is helpful because it shows how commands work and explains their options.